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FIT 3181 Deep Learning

Week 01: Unit Logistic

CE/Lecturer: Dr Trung Le (Clayton)

Lecturer: Dr Binh Nguyen (Clayton)

Lecturer: Dr Sicily Ting Fung (Malaysia)

Email: trunglm@monash.edu



GROUP OF EIGHT AUSTRALIA

Department of Data Science and AI
Faculty of Information Technology, Monash University, Australia


VEED

Outline

- Unit Logistics
- Mathematical Background Revisit
- Machine Learning Revisit
- Logistic Regression
- Summary

Unit Logistics

Unit logistics

FIT3181, 2026



Dr. Trung Le



Dr Binh Nguyen



Dr Sicily Ting Fung



Mr Minh Vo
[Head Tutor 3181]



Anh



Dongwon



Simin

- FIT 3181 Team
  - **Lecturers**: **Dr Trung Le** (trunglm@monash.edu), **Dr Binh Nguyen** (binh.nguyen1@monash.edu), and **Dr Sicily Ting Fung** (ting.fung@monash.edu)
  - **Head Tutor**: **Mr Minh Vo** (minh.vo@monash.edu)

Lecture and consultation times

- Lectures for FIT 5215 will be conducted online
  - Time for FIT5215: Monday 12:00 pm – 2:00 pm
  - Zoom link: https://monash.zoom.us/j/88033513605?pwd=CeKjxJegBzqFBDOpiTUrNlouusEp5a.1

  - Time for FIT3181: Friday 8:00 am – 10:00 am
  - Zoom link: https://monash.zoom.us/j/88033513605?pwd=CeKjxJegBzqFBDOpiTUrNlouusEp5a.1

- Important notes:
  - There will be 12 lectures in total, each is 2 hours long.
  - There will be 22 consultations during teaching (1 hour long).

Tutorials

- Make sure you have been allocated to one tutorial.
- **They are on-campus**
  - Make sure you know the tutorial room
- If you miss a tutorial and would like to join the next (same) tutorial, you must email the tutor for permission and detail to attend.

Assessments

Tests, Assignments and Exam

- **Two** in-semester online tests: **10% each**
  - During week **7** and week **12**
  - Approximately 1 hour long, online test, questions will be randomised each time.
- **Two** assignments: **25% (A1) and 20% (A2)**
  - (Tentatively) due in week **7** and week **12** (Friday)
- **One** final exam (closed book): **35%**
  - **10** minutes reading
  - Duration: **2 hours**
- To pass this unit:
  - 40% or more in the final exam
  - An overall unit mark of 50% or more

Assessments

Tests, Assignments and Exam

- Late submission
  - Subject to marking penalty
  - 5% deduction per day

- Apply for extension:
  - Applied for short extensions

Communication

- Contact your CE/lecturer, tutor for your concerns as soon as possible.
- Strongly recommend to use Ed discussion as much as possible!

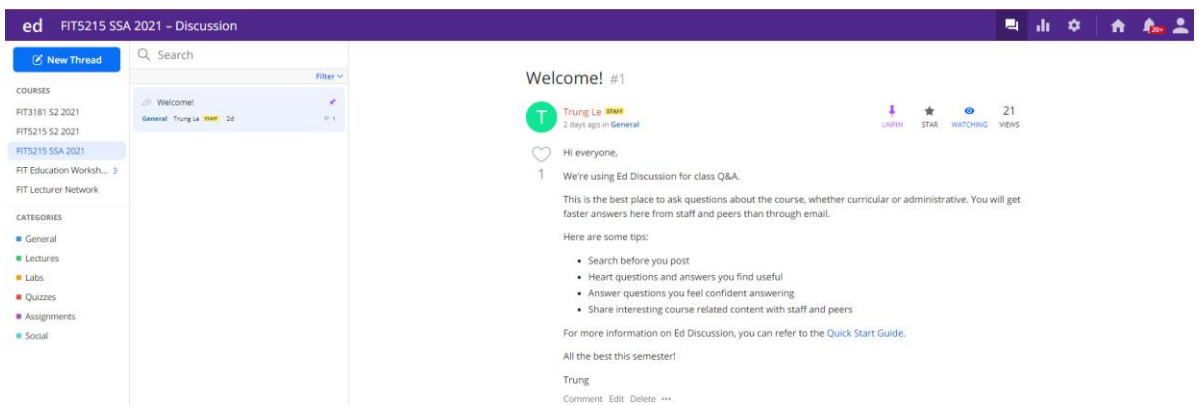
Forums:

Expectations for Forum Communication:

We will use Ed forum for announcements, discussions, questions, etc. Here are some resource

- Be **courteous**. Don't write anything you would not say face-to-face or that you would not war
- Be **helpful**, within limits. You can help each other, but do not give answers to assessed assig academic integrity
- Be **generous**. We encourage you "give" at least as much as you "take", to the extent that it is availability is limited. Having the cohort's help will make a day-and-night difference in everyc
- Be **clear**. Review what you write to make sure others will understand your message and inte strive twice as hard to be understood, as we do not have the benefit of modifying or elabora
- Be **self-sufficient**. Before asking questions, do your own research to ensure that the answer can post your questions with the answers you have found and ask for a second opinion.
- Monash takes its **student charter** seriously. Any posts deemed inappropriate by your teachi
- Please **review** the resources provided below to ensure that you are using all of ed's discussi

Ed Discussion



Learning objectives

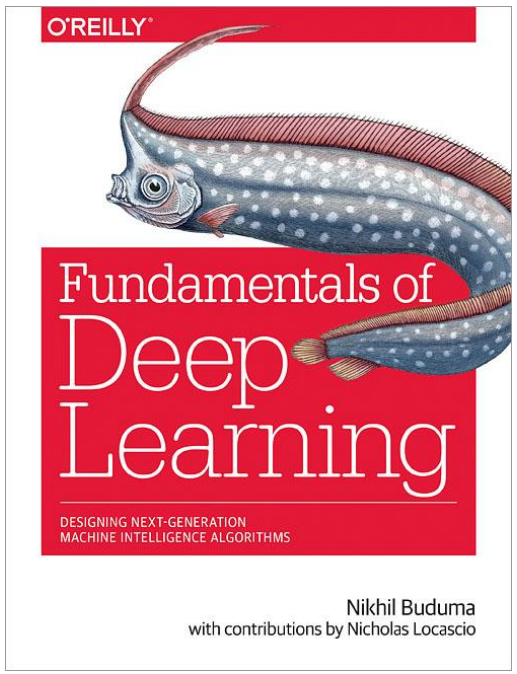
- Fundamental elements of deep learning
- Basic and advanced concepts of machine learning, AI and deep learning
- Hand-on experience with practical deep learning models and framework

Course structure

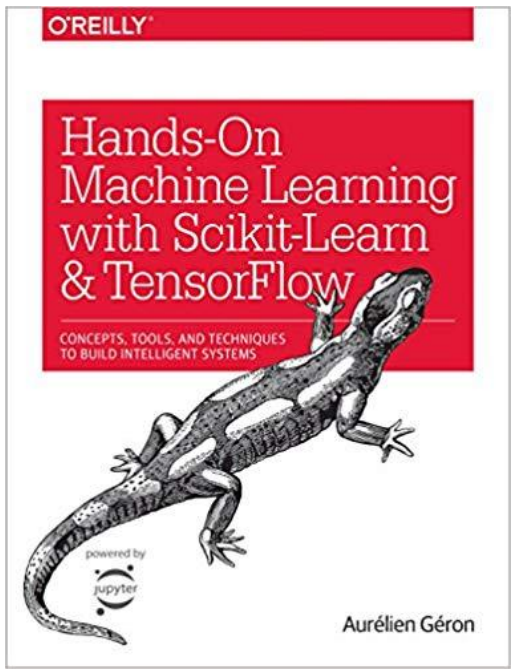
Week	Lectures	Assessment
1	Machine learning and mathematical background revisited	Assignment 1 released
2	Prelude to deep learning and feed-forward neural networks	
3	Deep Learning for Vision (I): convolutional neural networks (CNNs)	
4	Stochastic Gradient Descent and Optimization for Deep Learning	
5	Training (very) deep networks: optimizers and practical skills	
6	Deep Learning for Vision (II): network architectures, visualization, interpretability and robustness	
7	Deep learning for time-series and temporal data: RNNs and LSTMs/Exam Information	Online Quiz Test 1 (10%) Assignment 1 due (20%) Assignment 2 released
8	Neural embeddings: Word2Vec and Auto-encoders	
9	Seq2Seq and End-to-End Deep Models and Transformers	
10	Vision Transformers, Model Fine-tuning with Prompts, Adaptors, and LoRA	
11	Deep generative models: GANs and Diffusion Models	Online Quiz Test 2 (10%) Assignment 2 due (20%)
12	Revision and Exam Information	

Unit Logistics

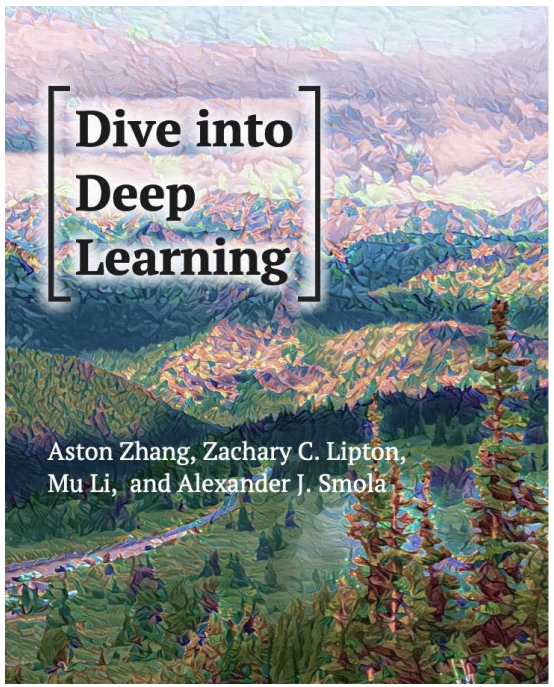
Textbook and References



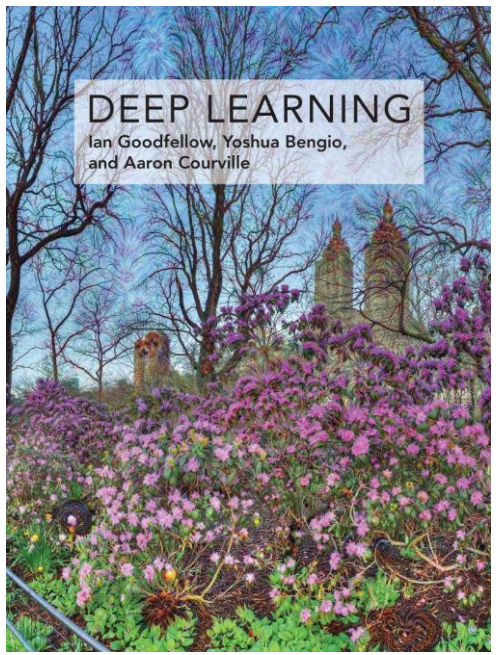
Introductory and hand-on



Intermediate and hand-on, both 1st and 2nd edition



Intermediate and hand-on (but don't use tensorflow)



Advanced and theoretical

- No single textbook, some tutorial/lectures parts might use these books
- All of these books are available as online resources from Monash library.

In-class Kaggle Competition

- This year, to increase **student engagement** in the labs/tutorials, we are applying a new teaching mode in which:
  - Students in each tutorial/lab class will **register and work in groups of 3 to 6**.
  - In several tutorials/labs, we will organize **in-class Kaggle competitions**. You and your group mates will rely on **the code provided in the tutorials/labs** to **develop and train specific deep learning models** and **submit the results to Kaggle** for your class.
  - If you **surpass our quite trivial baseline** ("**beat the first boss**"), you will win **1 star** (1 ★).
  - If you **surpass our second baseline** ("**beat the second boss**"), you will win **1 star** (2 ★).
  - If you are in **the top 1 of your class**, you will win 3 stars.
  - If you are in **the top 1 of entire cohort**, you will win 4 stars.
  - The stars you earn can be **converted to bonus points** in assignments, with **2 stars equaling 1 point**, up to a **maximum of 10 bonus points**. Any remaining stars will be **converted to extra time** for in-semester quizzes, with 1 star equaling 1 extra minute.
  - Data and tasks are sufficiently simple so that you can achieve good results in less than one hour. Additionally, in the first half of the tutorials/labs, we will introduce and discuss the lab/tutorial materials with you to provide a solid background for the Kaggle tasks.
- In the first tutorial/lab, please identify and connect with others to form your group.

Unit Logistics

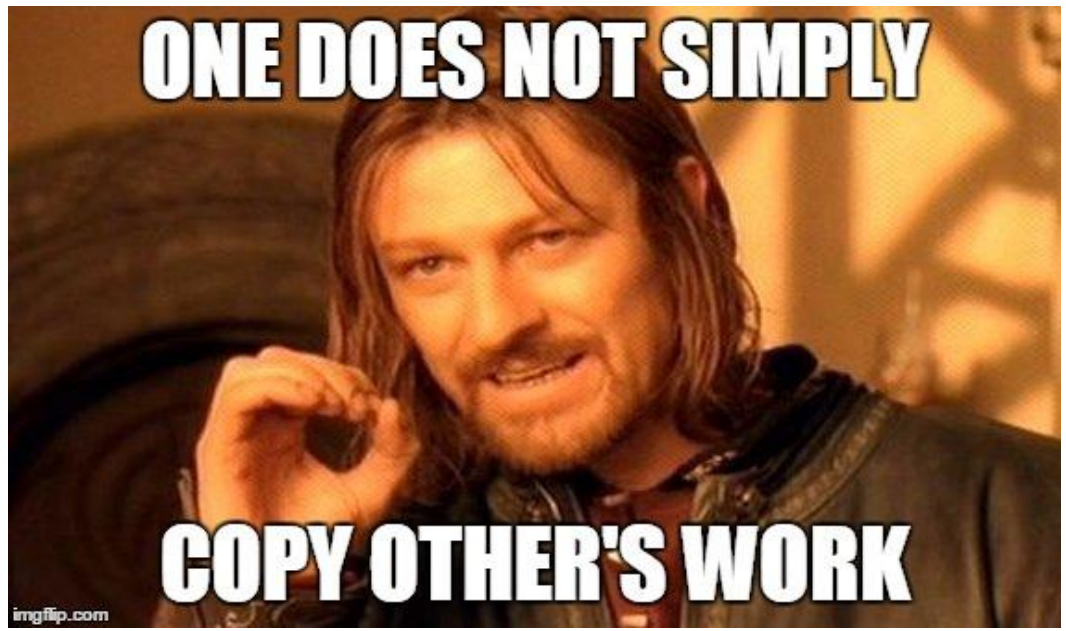
- How to succeed in this unit?
  - Work on the hand-on tutorials and master them as quickly as you can
  - Where applicable, attempt all exercises and questions during lectures and tutorials
  - Revise key points after each lesson to prepare for the in-semester test
  - Start on your assignments early!
  - Build your knowledge systematically from lectures and recommended readings
    - If it helps, use some mindmap tool such as xmind
- How to become extra … ordinary?
  - Read and learn beyond what is being taught
  - Work on project/ideas that inspire you
  - Be proactive, github and build DL profile

Plagiarism Notice

- Plagiarism:

  "the practice of taking someone else's work or ideas and passing them off as one's own" (Oxford Dictionary)

- It is a serious academic offence and you can be expelled from the University!




Q&A
When?
Why?
How?
WHAT?
WHERE?
WHEN?
HOW?
WHO?
What?
Where?